CSC 252 — Test 3 Study Guide (spaced / light)

Banner — scope PROVISIONAL / LIGHT. Test 3 scope is **not published**. The **date is confirmed**: Thu Oct 15, 2026, during lecture, 25 minutes. The syllabus lectures just before Test 3 are Deck 05 (The ALU), Deck 06 (MIPS Functions), and Deck 07 (Instruction Formats and CPU Details). The course site does **not** assign those decks to any test, so this guide does **not** claim them as scope. Until coverage is published, this guide keeps Decks 01–04 skills warm and offers an **optional, clearly labelled preview** (§4c) to use only after lecture covers those topics.

Canonical maps: `csc252-topic-map.md` · `announced-assessments-inventory-2026-09-24.md` · `site-audit-2026-09-24.md`

1. Header

Field	Value	Status
Course	CSC 252 Computer Organization (FA26), TuTh 9:30–10:45 AM	Confirmed
Assessment	Test 3	Confirmed
Date	**Thu Oct 15, 2026**	**Confirmed** (homepage)
Length / when	25 minutes during lecture; exact clock time **not published**	Length confirmed; clock time NV
Coverage	not published	**PROVISIONAL**
Guide updated	2026-09-24 (America/Phoenix) — Opus review pass	

All times are **Arizona time (MST)**.

Nearby deadlines (energy planning only)

Item	When (Arizona)	Status
Sim 3 ("An ALU")	Wed Oct 7, 7:00 PM	Homepage + Gradescope
HW3	Tue Oct 13, 7:00 PM	**Rule-derived** from "due two days before each test". Not individually posted yet.
Asm 3	about Wed Oct 14, 7:00 PM	Syllabus says "on or about" — approximate

2. Topic checklist (light / TBD-aware)

Order	Section ID	Topic	Status
1	`252-T3-tbd`	Official Test 3 topics	**TBD.** Watch the homepage, `tests/` folder, and lecture announcements.
2	`252-T3-cont-binary`	Continuation: binary, 2's complement, add/subtract	Keep warm. **Not** claimed as Test 3 scope.
3	`252-T3-cont-mips`	Continuation: MIPS loops and arrays	Keep warm. **Not** claimed as scope.
4	`252-T3-cont-gates`	Continuation: gates and adders	Keep warm. **Not** claimed as scope.
5	`252-T3-preview`	Optional preview of Decks 05–07 (§4c)	**Use only after lecture covers it.** Not claimed as scope.

When scope is published, replace row 1 and regenerate this guide.

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3. Explanations (light)

Keep this short. Until coverage is published:

- Do a 15-minute flash on Test 2 skills each week, so they stay warm for Sim 3 and later tests.
- After each lecture from Oct 1 on, write down 2–3 topic bullets and label each **"unverified until slides/homepage confirm."**
- Use the official deck titles, not guesses about what might be tested.

How the preview works. The deck titles near Test 3 are public. §4c turns them into self-check prompts based on **standard MIPS conventions** (Patterson & Hennessy). Match notation to the actual slides, and skip anything not yet lectured.

Sim 3 boundary. Sim 3 ("An ALU") prohibits AI help. §4c covers ALU **concepts only**. It contains no Sim 3 code, structure, or hints.

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4. Practice — continuity set (self-contained)

P1 (meta). What is confirmed about Test 3 today, and what is not?

P2 (binary add). Add 8-bit `0100 1011` + `0110 0110`. Give the result, whether there is unsigned overflow, and whether there is signed overflow.

P3 (2's complement). Negate `0001 1100` in 8-bit 2's complement. Then sign-extend the result to 16 bits, in hex.

P4 (subtraction). Compute `0x50 − 0x68` in 8 bits using "invert, add 1, add". Give the signed result and say whether a borrow happened.

P5 (MIPS loop). Write a loop that sums the word array at `$s0` with length `$s1` into `$s2`, using only allowable-style instructions.

P6 (gates). Write Cout of a full adder as a sum of products, and say how many full adders a 32-bit ripple-carry adder needs.

P7 (planning). After Oct 1, list three lecture topics that might feed Test 3. Label each **unverified**.

4b. Light Gradescope-format continuity — Practice recommendation

This is inferred from the Test 1 result structure only: a scanned, hand-written paper of about 6 pages, with short-concept, binary-arithmetic, and C-to-MIPS questions. It is **not** a published Test 3 format.

GS-T3-L1. Round-trip unsigned 8-bit `0xD4` to binary, then to decimal, then back to hex. Show work as on a test page.

GS-T3-L2. In one line each: is `0xD4` negative as 8-bit 2's complement? What is its signed value?

GS-T3-L3. Write 3–5 MIPS lines that clear `$t0`, load word 2 of the array at `$s0` into `$t1`, and add it to `$t0`.

4c. OPTIONAL preview — Decks 05–07 (not claimed scope)

Only use these after the matching lecture has happened. Answers use standard MIPS and Patterson & Hennessy conventions. If the slides use different notation, follow the slides.

PV1 (Deck 05, ALU concept). A 1-bit ALU slice computes AND, OR, and add, then picks one result. Which circuit element selects the result, and what drives its select lines?

PV2 (Deck 05, ALU concept). How can an adder-based ALU compute A − B without a separate subtractor circuit?

PV3 (Deck 06, functions). Which instruction calls a function, and where does it save the return address? Which instruction returns?

PV4 (Deck 06, functions). Which registers must a function preserve if it modifies them (callee-saved), and which may it overwrite freely (caller-saved)? Why does a function that calls another function need to save `$ra`?

PV5 (Deck 07, formats). Give the field layout and bit widths of the R-format, I-format, and J-format instructions.

5. ANSWER KEY

A1. Confirmed: Thu Oct 15, 2026, a 25-minute test during lecture. Not confirmed: scope and the exact clock time.

A2. `0100 1011` (75) + `0110 0110` (102) = `1011 0001` (177).

- No carry out, so **no unsigned overflow**.
- Both inputs are positive but the result's MSB is 1, so there **is signed overflow**. The true sum, 177, exceeds 127.

A3. `0001 1100` = 28. Inverted: `1110 0011`. Plus 1: `1110 0100` = −28 (0xE4). Sign-extended to 16 bits: `0xFFE4`.

A4. −`0x68`: `0110 1000` inverted is `1001 0111`, plus 1 is `1001 1000` (0x98). Then `0x50 + 0x98 = 0xE8` = `1110 1000`. As a signed value that is 232 − 256 = **−24** (check: 80 − 104 = −24). The carry-out is 0, so **a borrow happened**.

A5.

```
      add  $s2, $zero, $zero
      add  $t0, $zero, $zero
LOOP: beq  $t0, $s1, DONE
      sll  $t1, $t0, 2
      add  $t1, $s0, $t1
      lw   $t2, 0($t1)
      add  $s2, $s2, $t2
      addi $t0, $t0, 1
      j    LOOP
DONE:
```

A6. Cout = A·B + A·Cin + B·Cin. A 32-bit ripple-carry adder needs 32 full adders. Bit 0 can be a half adder if there is no carry-in, but using a full adder there enables subtraction by setting Cin = 1.

A7. Your own list, with every item labelled unverified.

GS-T3-L1. `0xD4` = `1101 0100` = 128 + 64 + 16 + 4 = 212, which is back to `0xD4`.

GS-T3-L2. Yes, it is negative (MSB 1). 212 − 256 = **−44**.

GS-T3-L3.

```
add  $t0, $zero, $zero     # t0 = 0
lw   $t1, 8($s0)           # word index 2 -> byte offset 8
add  $t0, $t0, $t1
```

Preview keys (confirm against slides)

PV1. A multiplexer (MUX) selects the result. The operation / control bits drive its select lines.

PV2. Invert B (XOR each bit with 1, or use a "B-invert" MUX) and set the carry-in to 1. That computes A + (~B) + 1 = A − B, which is 2's complement again.

PV3. `jal label` jumps to the function and saves the return address (PC + 4) in `$ra`. `jr $ra` returns.

PV4. Callee-saved: `$s0–$s7` (plus `$sp`, and `$ra` in practice). Caller-saved: `$t0–$t9`, `$a0–$a3`, `$v0–$v1`. A function that makes a call runs `jal`, which overwrites `$ra`, so it must push `$ra` on the stack first and restore it before `jr $ra`.

PV5. Each format totals 32 bits.

Format	Fields (bit widths)
R-format	op (6), rs (5), rt (5), rd (5), shamt (5), funct (6)
I-format	op (6), rs (5), rt (5), immediate (16)
J-format	op (6), address (26)

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6. Common mistakes + readiness

- Treating Test 2 topics, or the §4c preview, as confirmed Test 3 scope.
- Studying Deck 05–07 material before it has been lectured, or instead of Sim 3.
- Mixing up unsigned overflow (carry-out) with signed overflow (sign flip).
- Skipping sleep the night before Oct 15.

Readiness (spaced): 2–3 short sessions per week after Oct 8. Expand once scope is published or decks have been lectured.

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7. Day-by-day (spaced; no day-before cram)

Weekdays checked against the 2026 calendar.

Day	Plan
Fri Oct 2 – Wed Oct 7	Almost nothing for Test 3. Finish Sim 3 (due Wed Oct 7, 7:00 PM). Optional 15-minute continuity flash.
Thu Oct 8 – Sat Oct 10	Light: add lecture notes to `252-T3-tbd`; one short MIPS or binary flash.
Sun Oct 11	Short mixed continuity set, plus §4c for any lectured decks.
Mon Oct 12	Structured review of **known, lectured** material only. Deep-ish day.
Tue Oct 13	**LIGHT** only — class day; HW3 due 7:00 PM (rule-derived).
Wed Oct 14	Calm recall — **no cram**. Asm 3 due about 7:00 PM (approximate).
Thu Oct 15 morning	Buffer, then Test 3 in lecture.

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8. Changes in this review (2026-09-24)

- Fixed the day plan's weekday errors. The old plan said "Mon Oct 13" and "Tue Oct 14" and had a stray "Wed Oct 14? wait" row; in 2026, Oct 12 is Monday, Oct 13 Tuesday, and Oct 14 Wednesday.
- Made the answer keys self-contained. The old keys said "see the Test 2 guide"; A2–A4 and L1–L3 now have full worked answers.
- Added the 25-minute length, clock-time-NV status, and nearby deadlines. HW3 is labelled **rule-derived**, and Asm 3 is labelled **approximate**.
- Added the optional §4c preview for Decks 05–07. It uses standard MIPS conventions, is explicitly **not** claimed as scope, is gated on lecture coverage, and gives no Sim 3 hints.
- Changed times to Arizona time.

9. Disclaimer

Study aid only — not for submitted work. Follow CSC 252 and UA academic-integrity and AI policies. Coverage is **provisional/TBD** until the instructor publishes it.